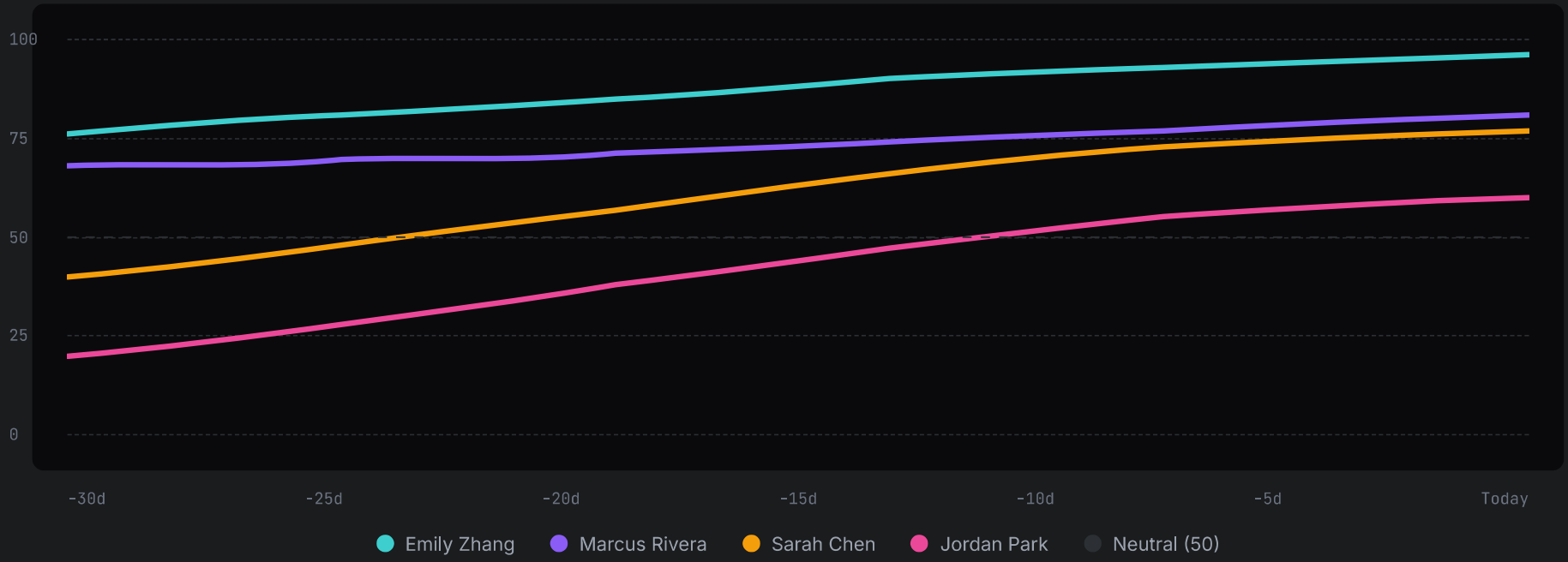
PDE Time Decay Analysis

Dashboard → Analytics → Cultural Momentum

KQ Decay Trajectories

$$q(t+1) = \rho \cdot q(t) + (1-\rho) \cdot s(t)$$

where $\rho = \exp(-\gamma \cdot \lambda_max)$ controls decay rate based on principal eigenvalue



0.92	0.055	1.52	30d
ρ (Decay Factor)	γ (Decay Rate)	λ_max (Eigenvalue)	Window Size

👥 Individual Trajectories

TEAM MEMBER	CURRENT KQ	30D Δ	MOMENTUM	TRAJECTORY
E Emily Zhang Engineering	98	↑ +12	Strong positive	Rising leader – consistent recognition behavior
M Marcus Rivera Product	94	↑ +4	Stable high	Maintaining excellent culture contribution
S Sarah Chen Design	89	↑ +18	Accelerating	Rapid improvement – cross-team collaboration up
J Jordan Park Engineering	78	↑ +8	Recovering	Bouncing back after project crunch period

🔮 Predictive Insight

Based on current trajectories, team average KQ is projected to reach **74** within 14 days if recognition patterns continue. Sarah's acceleration suggests potential for top-5 ranking by month-end. Consider nudging Jordan with a recognition prompt to accelerate recovery.